

# Postal MS

## Sprint Planning Document

4 Sprints | 3 March 2026 - 19 April 2026 | 71 Tasks Total | All Completed

### Project Overview

PostalMS was developed using the SCRUM agile methodology across four sprints. Sprint planning was conducted during team meetings held in the Ritterman Building and Sheppard Library at Middlesex University.

### Team Members

| Name                  | Student ID | Role                                     |
|-----------------------|------------|------------------------------------------|
| Durga Prasad Avala    | M01038074  | Team Leader and Developer (SCRUM Master) |
| Meet Sanjay Mande     | M01000862  | Secretary                                |
| Tapan Keyurbhai Patel | M01050963  | Developer                                |
| Forhad Hossain        | M01044942  | Tester                                   |

### Priority Key

| Priority | Description                                                                      |
|----------|----------------------------------------------------------------------------------|
| High     | Must be completed for the application to function or for submission requirements |
| Medium   | Important but can be completed after High priority tasks                         |
| Low      | Administrative tasks that support the project                                    |

# Sprint 1 -- Project Setup and Data Structure Design

|                      |                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Sprint Number</b> | Sprint 1 of 4                                                                                                  |
| <b>Start Date</b>    | Monday 3 March 2026                                                                                            |
| <b>End Date</b>      | Sunday 9 March 2026                                                                                            |
| <b>Sprint Goal</b>   | Set up the GitHub repository, create the initial project structure and design all three custom data structures |
| <b>Total Tasks</b>   | 10                                                                                                             |
| <b>Completed</b>     | 10                                                                                                             |
| <b>Status</b>        | Completed                                                                                                      |

| ID           | Task                                                            | Assigned To | Priority | Status    |
|--------------|-----------------------------------------------------------------|-------------|----------|-----------|
| <b>S1-01</b> | Create GitHub repository with correct folder structure          | Durga       | High     | Completed |
| <b>S1-02</b> | Set up C# WinForms project and push initial solution to repo    | Durga       | High     | Completed |
| <b>S1-03</b> | Design CustomHashTable class with chaining collision resolution | Durga       | High     | Completed |
| <b>S1-04</b> | Design CustomBST class with recursive insert and search         | Durga       | High     | Completed |
| <b>S1-05</b> | Research and plan CustomQueue linked list implementation        | Tapan       | High     | Completed |
| <b>S1-06</b> | Create class diagram in draw.io and push to design folder       | Meet        | Medium   | Completed |
| <b>S1-07</b> | Create ERD diagram in draw.io and push to design folder         | Meet        | Medium   | Completed |
| <b>S1-08</b> | Set up MSTest project referencing main application              | Forhad      | Medium   | Completed |
| <b>S1-09</b> | Review CST2550 coursework brief and identify all requirements   | All         | High     | Completed |
| <b>S1-10</b> | Write Sprint 1 meeting minutes and distribute to team           | Meet        | Low      | Completed |

## Sprint 1 Review

- All 10 tasks were completed. The GitHub repository was created, the project structure was set up, and all three custom data structures were designed and implemented.

## Sprint 1 Retrospective

- What went well: Team roles were assigned clearly, and the repository was set up quickly.
- What could be improved: More detailed pseudocode should have been written before starting implementation.

## Sprint 2 -- Database and Core UI

|                      |                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------|
| <b>Sprint Number</b> | Sprint 2 of 4                                                                                         |
| <b>Start Date</b>    | Monday 10 March 2026                                                                                  |
| <b>End Date</b>      | Sunday 23 March 2026                                                                                  |
| <b>Sprint Goal</b>   | Build the SQL Server database, implement login and registration, and create the main navigation shell |
| <b>Total Tasks</b>   | 14                                                                                                    |
| <b>Completed</b>     | 14                                                                                                    |
| <b>Status</b>        | Completed                                                                                             |

| ID    | Task                                                                | Assigned To | Priority | Status    |
|-------|---------------------------------------------------------------------|-------------|----------|-----------|
| S2-01 | Create SQL Server database with Users, Parcel and Delivery tables   | Durga       | High     | Completed |
| S2-02 | Write all stored procedures for CRUD operations                     | Durga       | High     | Completed |
| S2-03 | Implement DatabaseHelper.cs with connection and query methods       | Durga       | High     | Completed |
| S2-04 | Implement LoginForm with Gmail validation and session management    | Durga       | High     | Completed |
| S2-05 | Implement RegisterForm with all user fields and validation          | Durga       | High     | Completed |
| S2-06 | Implement MainForm with sidebar navigation and panel switching      | Durga       | High     | Completed |
| S2-07 | Implement DataStructuresView showing live Hash Table, BST and Queue | Tapan       | High     | Completed |
| S2-08 | Write unit tests for Hash Table operations (12 tests)               | Forhad      | High     | Completed |
| S2-09 | Write unit tests for Queue operations (10 tests)                    | Forhad      | High     | Completed |
| S2-10 | Create pseudocode document for Hash Table and Login logic           | Meet        | Medium   | Completed |
| S2-11 | Create time complexity analysis document for all three structures   | Meet        | High     | Completed |
| S2-12 | Test login and registration flows end to end                        | Forhad      | High     | Completed |
| S2-13 | Write Sprint 2 meeting minutes                                      | Meet        | Low      | Completed |
| S2-14 | Push all Sprint 2 code to GitHub                                    | Durga       | High     | Completed |

### Sprint 2 Review

- All 14 tasks were completed. The database was created with all tables and stored procedures. Login, registration and main navigation were implemented and tested.

### Sprint 2 Retrospective

- What went well: Database setup was smooth and DatabaseHelper made queries easy across all views.

- What could be improved: UI styling should have been standardised earlier to avoid inconsistencies.

## Sprint 3 -- Full Application and Testing

|                      |                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------|
| <b>Sprint Number</b> | Sprint 3 of 4                                                                                  |
| <b>Start Date</b>    | Monday 24 March 2026                                                                           |
| <b>End Date</b>      | Thursday 9 April 2026                                                                          |
| <b>Sprint Goal</b>   | Complete all remaining WinForms views, write all unit tests, complete all documents and submit |
| <b>Total Tasks</b>   | 21                                                                                             |
| <b>Completed</b>     | 21                                                                                             |
| <b>Status</b>        | Completed                                                                                      |

| ID    | Task                                                                | Assigned To | Priority | Status    |
|-------|---------------------------------------------------------------------|-------------|----------|-----------|
| S3-01 | Implement ParcelsView with send form and parcel list                | Durga       | High     | Completed |
| S3-02 | Implement DeliveriesView with tracking timeline and driver details  | Durga       | High     | Completed |
| S3-03 | Implement HomeView, HelpView and InfoView pages                     | Durga       | Medium   | Completed |
| S3-04 | Implement ProfileView with user details display                     | Durga       | Medium   | Completed |
| S3-05 | Implement offline AI assistant with parcel tracking and price logic | Durga       | High     | Completed |
| S3-06 | Write unit tests for BST operations (13 tests)                      | Forhad      | High     | Completed |
| S3-07 | Write unit tests for price calculation (9 tests)                    | Forhad      | High     | Completed |
| S3-08 | Write unit tests for tracking ID validation (6 tests)               | Forhad      | High     | Completed |
| S3-09 | Write integration tests for all three data structures (5 tests)     | Forhad      | High     | Completed |
| S3-10 | Run all 59 unit tests and confirm all pass in Visual Studio         | Forhad      | High     | Completed |
| S3-11 | Complete pseudocode for BST and Queue with time complexity tables   | Meet        | High     | Completed |
| S3-12 | Complete PDF report design, testing and conclusion sections         | Meet        | High     | Completed |
| S3-13 | Assist with UI form layouts and final styling adjustments           | Tapan       | Medium   | Completed |
| S3-14 | Record MP4 video demonstration under 6 minutes                      | Tapan       | High     | Completed |
| S3-15 | Final check of all repo folders and confirm everything is uploaded  | Forhad      | High     | Completed |

| ID    | Task                                                         | Assigned To | Priority | Status    |
|-------|--------------------------------------------------------------|-------------|----------|-----------|
| S3-16 | Push all source code and test project to GitHub              | Durga       | High     | Completed |
| S3-17 | Invite all 4 markers to GitHub repository as collaborators   | Durga       | High     | Completed |
| S3-18 | Document Meeting 3 and Meeting 4 minutes and upload to repo  | Meet        | Low      | Completed |
| S3-19 | Submit .txt file with GitHub URL before deadline             | Durga       | High     | Completed |
| S3-20 | Add drop off locations with Google Maps links to ParcelsView | Durga       | Medium   | Completed |
| S3-21 | Implement FindUsView with PostalMS office locations          | Durga       | High     | Completed |

### Sprint 3 Review

- All 21 tasks were completed successfully. All WinForms views were implemented, all 59-unit tests passed, and all documents were completed and submitted before the deadline.

### Sprint 3 Retrospective

- What went well: All 59-unit tests passed on the first run with no failures. The AI assistant bonus feature was completed successfully.
- What could be improved: Report writing could have started earlier to reduce time pressure.

## Sprint 4 -- Feature Enhancements and Final Improvements

|                      |                                                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sprint Number</b> | Sprint 4 of 4                                                                                                                                                                                  |
| <b>Start Date</b>    | Friday 10 April 2026                                                                                                                                                                           |
| <b>End Date</b>      | Friday 17 April 2026                                                                                                                                                                           |
| <b>Sprint Goal</b>   | Implement professor-requested improvements: stamps ordering, Find Us city detection, Gmail validation, full receiver details, DOB in collect, drop off maps, ProfileView fix and 78-unit tests |
| <b>Total Tasks</b>   | 26                                                                                                                                                                                             |
| <b>Completed</b>     | 26                                                                                                                                                                                             |
| <b>Status</b>        | Completed                                                                                                                                                                                      |

| ID    | Task                                                                                 | Assigned To | Priority | Status    |
|-------|--------------------------------------------------------------------------------------|-------------|----------|-----------|
| S4-01 | Remove stamps tab and Buy Stamps button from ParcelsView completely                  | Durga       | High     | Completed |
| S4-02 | Build complete StampsView with Browse, Order and My Orders tabs                      | Durga       | High     | Completed |
| S4-03 | Add Delivery vs Collect card selection (food-app style) to stamp ordering            | Durga       | High     | Completed |
| S4-04 | Add full delivery details: name, address, city, postcode, email, delivery speed      | Durga       | High     | Completed |
| S4-05 | Add full collect details: name, phone, DOB, location, date, time slot, email         | Durga       | High     | Completed |
| S4-06 | Add Pay Now Online vs Pay In Store card selection to stamp ordering                  | Durga       | High     | Completed |
| S4-07 | Add stamp price breakdown: stamps + delivery + 20% VAT + 2% service fee              | Durga       | High     | Completed |
| S4-08 | Add missed pickup logic: 48 hour hold, auto-reschedule and red banner                | Durga       | Medium   | Completed |
| S4-09 | Expand ParcelsView Step 4: name, phone, email, address, city, postcode, instructions | Durga       | High     | Completed |
| S4-10 | Add UK city dropdown for domestic destination in ParcelsView                         | Durga       | High     | Completed |
| S4-11 | Fix international country dropdown overlap with Step 4 in ParcelsView                | Durga       | High     | Completed |
| S4-12 | Add drop off locations with Google Maps links in ParcelsView                         | Durga       | Medium   | Completed |
| S4-13 | Rewrite FindUsView with 7 cities and 10 plus locations per city                      | Durga       | High     | Completed |
| S4-14 | Add city auto-detection from user profile and city dropdown in FindUsView            | Durga       | High     | Completed |
| S4-15 | Add City field to RegisterForm and City column to Users database table               | Durga       | High     | Completed |
| S4-16 | Update RegisterUser stored procedure to accept and store city parameter              | Durga       | High     | Completed |

| ID    | Task                                                                                     | Assigned To | Priority | Status    |
|-------|------------------------------------------------------------------------------------------|-------------|----------|-----------|
| S4-17 | Fix ProfileView blank page bug using Q_Public direct database query                      | Durga       | High     | Completed |
| S4-18 | Enforce Gmail validation in LoginForm, RegisterForm, StampsView and ParcelsView          | Durga       | High     | Completed |
| S4-19 | Write 19 new tests: GmailValidationTests (8), StampPriceTests (6), CityLocationTests (8) | Forhad      | High     | Completed |
| S4-20 | Make test file self-contained with no external project reference needed                  | Forhad      | High     | Completed |
| S4-21 | Run all 78 tests and confirm 0 failures                                                  | Forhad      | High     | Completed |
| S4-22 | Update Unit Testing Report to cover all 78 tests across 9 test classes                   | Meet        | High     | Completed |
| S4-23 | Update Sprint Planning document to add Sprint 4 with all 26 tasks                        | Meet        | High     | Completed |
| S4-24 | Update Pseudocode document with Gmail, stamps, city and receiver algorithms              | Meet        | High     | Completed |
| S4-25 | Update Time Complexity document with Sprint 4 algorithm analysis                         | Meet        | High     | Completed |
| S4-26 | Push all Sprint 4 changes to GitHub                                                      | Durga       | High     | Completed |

## Sprint 4 Review

- All 26 tasks were completed successfully. The stamps ordering system was built from scratch as a completely separate page with Delivery vs Collect card selection, Pay Now vs Pay In Store, full user details including DOB, and missed pickup logic. The Find Us page was rewritten with 7 cities and 76 locations. Gmail validation was enforced throughout. The ProfileView blank page bug was fixed. All 78 unit tests pass with 0 failures.

## Sprint 4 Retrospective

- What went well: The stamps ordering system was implemented cleanly. All 78 tests pass with 0 failures. City-based location detection works automatically from the user profile.
- What could be improved: Better initial UI layout planning would have reduced overlap fix iterations. Test file architecture should have been self-contained from the start.
- Action for future: Plan UI Y positions with fixed spacing from the beginning to avoid overlapping controls.

## Project Summary

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| Metric                   | Value                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------|
| Total Sprints            | 4                                                                                          |
| Total Tasks              | 71 (10 + 14 + 21 + 26)                                                                     |
| Tasks Completed          | 71                                                                                         |
| Tasks Incomplete         | 0                                                                                          |
| Unit Tests Written       | 78                                                                                         |
| Unit Tests Passed        | 78                                                                                         |
| Unit Tests Failed        | 0                                                                                          |
| Test Classes             | 9 (HashTable, BST, Queue, Price, TrackingID, Integration, Gmail, StampPrice, CityLocation) |
| Cities in Find Us        | 7 (London, Manchester, Birmingham, Leeds, Bristol, Edinburgh, Glasgow)                     |
| Total PostalMS Locations | 76 locations across all 7 cities                                                           |
| Project Start            | Monday 3 March 2026                                                                        |
| Project End              | Friday 17 April 2026                                                                       |

- All 71 tasks across all 4 sprints were completed successfully. PostalMS was built using C# WinForms and SQL Server with three custom data structures. The stamps ordering system, city-based Find Us page, Gmail validation, full receiver details with DOB in collect, drop off Google Maps links and ProfileView bug fix were all added in Sprint 4. All 78 unit tests pass with 0 failures.